

Performance Testing

Date	7 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / Locust
Type of Testing	Load Testing, Stress Testing
Target Module / API	/ask, /quiz, and /summarize API Endpoints
Test Environment	
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate answers for multiple student queries simultaneously (/ask).	10	60	Contextual answers are generated successfully from the Gemini API without dropped tokens.
2	Concurrent quiz generation requests for "Pythagoras Theorem" (/quiz).	20	60	The backend system handles concurrent requests and delivers structured quiz objects without system failure.
3	Multiple users generating local text summaries simultaneously (/summarize).30	30	120	Response execution time remains stable, and the local LaMini-Flan-T5 model maintains stability under load.
4	Continuous structural learning	50	180	The API layer remains

	roadmap requests to the /roadmap endpoint.			stable with minimal errors, managing asynchronous cloud worker requests efficiently.
--	--	--	--	--



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Delay due to AI inference
3	Throughput (Req/sec)		12 req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.4%	Pass	Few timeout errors observed
5	CPU Utilization	< 80%	65%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limits

Step 4: Observations & Analysis

The core educational endpoints successfully handled concurrent requests without significant degradation. Average response time remained below the target threshold. Spikes in maximum response times occurred during peak load but remained within acceptable limits due to the network round-trip time of external cloud AI inference.

Key Findings:
[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
[List any performance bottlenecks observed during testing]

Optimization Steps Taken:
[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]